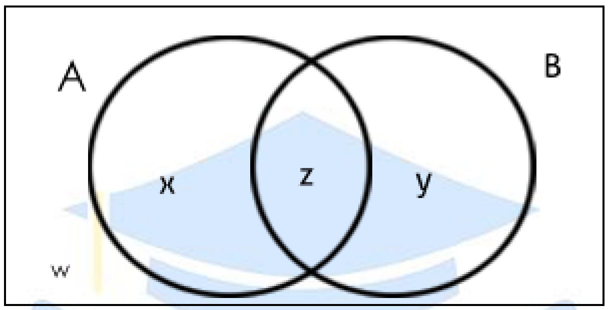
Venn Diagram

Venn diagram, also known as Euler-Venn diagram is a simple representation of sets by diagrams. It shows **logical relationships** between two or more sets (grouping items).The usual depiction makes use of a rectangle as the universal set and circles(both overlapping and nonoverlapping) for the sets under consideration.

In entrance exams, questions asked from this topic involve 2 or 3 variables only. Therefore, in this article we are going to discuss problems related to 2 and 3 variables.

Venn Diagram in case of two elements



Let's take a look at some basic formulas for Venn diagrams of two variables/elements/sets.

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

And so on, where $n(A)$ = number of elements in set A.

Once you understand the concept of Venn diagram with the help of diagrams, you don't have to memorize these formulas. Where;

X = number of elements that belong to set A only

Y = number of elements that belong to set B only

Z = number of elements that belong to set A and B both ($A \cap B$)

W = number of elements that belong to none of the sets A or B

From the above figure, it is clear that

$n(A) = x + z$;

$n(B) = y + z$;

$n(A \cap B) = z$;

$n(A \cup B) = x + y + z$.

Total number of elements = $x + y + z + w$

Tip: Always start filling values in the Venn diagram from the innermost value.

union <----> (or)

intersection <----> (and)

SAMPLE PROBLEMS

1. In a town, 800 people are selected by random types of sampling methods. 280 go to work by car only, 220 go to work by bicycle only and 140 use both ways – sometimes go with a car and sometimes with a bicycle. Find the answer of following questions:

- How many people go to work by car only?
- How many people go to work by bicycle only?
- How many people go by neither car nor bicycle?
- How many people use at least one of both transportation types?
- How many people use only one of car or bicycle?

2. In a college, 200 students are randomly selected. 140 like tea, 120 like coffee and 80 like both tea and coffee. Find the answer of following questions:

- How many students like only tea?
- How many students like only coffee?
- How many students like neither tea nor coffee?
- How many students like only one of tea or coffee?
- How many students like at least one of the beverages?

3. A travel agent surveyed 100 people to find out how many of them had visited the cities of Melbourne and Brisbane. Thirty-one people had visited Melbourne, 26 people had been to Brisbane, and 12 people had visited both cities. Draw a Venn diagram to find

- the number of people who had visited Melbourne or Brisbane
- the number of people who had visited Brisbane but not Melbourne
- the number of people who had visited only one of the two cities
- the number of people who had visited neither city.

4. At Adan's Automotive Shop, 50 cars were inspected. 23 of the cars needed new brakes, 34 needed new exhaust systems, and 6 cars needed neither repair. Draw a Venn diagram to find

- How many cars needed both repairs?
- How many cars needed new brakes, but not a new exhaust system?

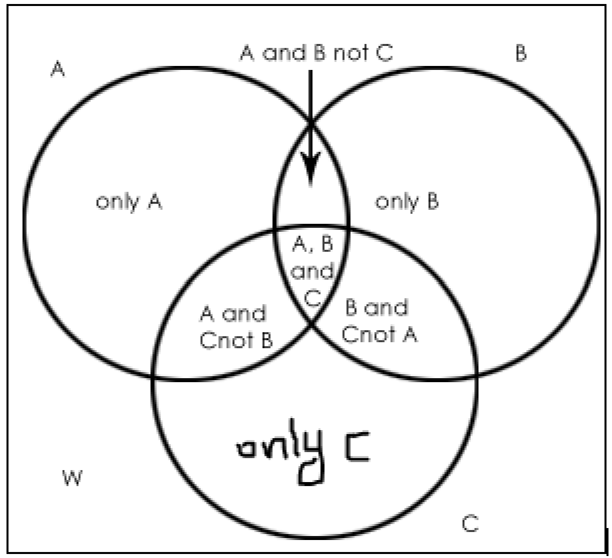
5. A survey of 64 informed voters revealed the following information:
   45 believe that Elvis is still alive.
   49 believe that they have been abducted by space aliens
   42 believe both of these things 1.

- How many believe neither of these things?
- How many believe Elvis is still alive but don't believe that they have been abducted by space aliens?

6. A survey of used car salesmen revealed the following information:
   24 wear white patent-leather shoes
   28 wear plaid trousers 20 wear both of these things
   2 wears neither of these things

- How many were surveyed?
- How many wear plaid trousers but don't wear white patent-leather shoes?

Venn Diagram in case of three elements



Let's take a look at some basic formulas for Venn diagrams of three variables/elements/sets.

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

W = number of elements that belong to none of the sets A, B or C

Tip: Always start filling values in the Venn diagram from the innermost value.

union <----> (or)
intersection <----> (and)

SAMPLE PROBLEMS

1. Out of 50 students who exercise regularly, 25 jog, 20 play basketball and 15 swim. 10 play basketball and jog, 5 play basketball and swim, 7 jog and swim and 2 people do all three. How many students do not do any of these activities?
2. A survey of 85 students asked them about the subjects they liked to study. Thirty-five students liked math, 37 liked history, and 26 liked physics. Twenty liked math and history, 14 liked math and physics and 3 liked history and physics. Two students liked all three subjects.

- How many of these students like math or physics?
- How many of these students didn't like any of the three subjects?
- How many of these students liked math and history but not physics?

3. In a certain school, there are 180 pupils in Year 7. One hundred and ten pupils study French, 88 study German and 65 study Indonesian. Forty pupils' study both French and German, 38 study German and German only. Find the number of pupils who study:

- all three languages
- Indonesian only
- none of the languages
- at least one language
- either one or two of the three languages.

4. A survey of faculty and graduate students at the University of Florida's film school revealed the following information:

51 admire Moe	49 admire Larry
60 admire Curly	34 admire Moe and Larry
32 admire Larry and Curly	36 admire Moe and Curly
24 admire all three of the Stooges	1 admires none of the Three Stooges

- How many people were surveyed?

- How many admire Curly, but not Larry nor Moe?
- How many admire Larry or Curly?
- How many admire exactly one of the Stooges?
- How many admire exactly two of the Stooges?